

Finite-chain fluctuation relations for the boundary-driven resonant NLS chain

A proof ledger separating exact identities from open spectral assumptions

Working note

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Abstract

We analyze the continuous-time, projection-free Cartesian diffusion used by the entropy-production simulations of the boundary-driven resonant nonlinear Schrödinger chain. For every fixed finite chain length we prove coercivity of the physical energy, global nonexplosion, equal-temperature Gibbs invariance, the bath-heat formula, and an exact formal-adjoint symmetry of the medium-entropy tilted generator. The latter is

$$\mathcal{L}_{n,k}^* = \mathcal{T} \mathcal{L}_{n,1-k} \mathcal{T}$$

on compactly supported smooth functions. It implies the Gallavotti–Cohen symmetry of any principal scaled cumulant generating function admitted by the closed Feynman–Kac generator. We state separately the unresolved analytic steps: existence and uniqueness of the unequal-temperature steady state, the relevant principal-eigenvalue theorem on the unbounded state space, and existence of a nontrivial thermodynamic limit. Thus this note supplies a model-specific algebraic theorem and a precise proof program; it does not disguise the remaining spectral assumptions as completed results.

1 Frozen continuous-time model

Let $n \geq 2$, $c_j = x_j + iy_j$, $I_j = |c_j|^2$, and $M = \sum_{j=1}^n I_j$. The physical energy used by the Cartesian programs is

$$E(c) = \frac{1}{2}M^2 - \frac{1}{4} \sum_{j=1}^n I_j^2 + \sum_{j=2}^n \operatorname{Re} \left(c_j^2 \overline{c_{j-1}^2} \right). \quad (1)$$

It equals one half of the Hamiltonian denoted by H in the manuscript. Put $z = (x_1, y_1, \dots, x_n, y_n) \in \mathbb{R}^{2n}$ and $F_j = (\partial_{x_j} E, \partial_{y_j} E)$. Let J be the block-diagonal symplectic matrix with blocks $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and let P_r project onto the two real coordinates of site r . The continuous diffusion is

$$dz_t = J \nabla E(z_t) dt - \gamma \sum_{r \in \{1, n\}} P_r \nabla E(z_t) dt + \sum_{r \in \{1, n\}} \sqrt{2\gamma T_r} P_r dW_{r,t}, \quad (2)$$

where the two boundary Brownian motions are independent. Its backward generator on smooth test functions is

$$\mathcal{L} = A + \sum_{r \in \{1, n\}} B_r, \quad A = (J \nabla E) \cdot \nabla, \quad B_r = -\gamma F_r \cdot \nabla_r + \gamma T_r \Delta_r. \quad (3)$$

The time-reversal involution is complex conjugation,

$$\Theta z = (x_1, -y_1, \dots, x_n, -y_n), \quad (\mathcal{T}f)(z) = f(\Theta z). \quad (4)$$

The energy is even, $E \circ \Theta = E$, and

$$\mathcal{T}A\mathcal{T} = -A, \quad \mathcal{T}B_r\mathcal{T} = B_r. \quad (5)$$

These identities fix the parity convention used below.

Remark 1.1 (Action–angle sign audit). Transforming the Cartesian bath in (2) to action–angle variables gives

$$b_1^\varphi = -2\gamma I_2 \sin \theta_2, \quad b_n^\varphi = +2\gamma I_{n-1} \sin \theta_n.$$

The advisor-synchronized manuscript snapshot has these signs. The enhanced working draft currently displays a plus sign at the left endpoint; that is a manuscript typo, not the Cartesian model analyzed here.

2 Energy geometry and global well-posedness

Lemma 2.1 (Coercivity). *For every $z \in \mathbb{R}^{2n}$,*

$$E(z) \geq \frac{1}{4} \sum_{j=1}^n I_j^2 \geq \frac{M^2}{4n} = \frac{|z|^4}{4n}. \quad (6)$$

Consequently E has compact sublevel sets and $\exp(-E/T)$ is integrable for every $T > 0$.

Proof. Writing $c_j = \sqrt{I_j} e^{i\varphi_j}$ and $\theta_j = 2(\varphi_j - \varphi_{j-1})$, expand (1) as

$$E = \frac{1}{4} \sum_j I_j^2 + \sum_{\substack{j < \ell \\ |j-\ell| > 1}} I_j I_\ell + \sum_{j=2}^n I_{j-1} I_j (1 + \cos \theta_j).$$

Every term after the first sum is nonnegative. The second inequality in (6) is Cauchy–Schwarz. $\square$

Lemma 2.2 (Boundary Laplacian and energy drift). *For each boundary site $r \in \{1, n\}$,*

$$\Delta_r E = 4M. \quad (7)$$

Moreover

$$\mathcal{L}E = -\gamma \sum_{r \in \{1, n\}} |F_r|^2 + 4\gamma(T_1 + T_n)M. \quad (8)$$

Proof. The interaction term is harmonic in each boundary pair (x_r, y_r) . Differentiating the first two terms of (1) gives $\Delta_r E = 4(M + I_r) - 4I_r = 4M$. The Hamiltonian vector field annihilates E , and applying the two bath generators gives (8). $\square$

Theorem 2.3 (Finite-chain nonexplosion). *For every fixed $n \geq 2$, positive temperatures, and deterministic initial condition, (2) has a unique global strong solution. In particular, the projection-free Cartesian process is nonexplosive.*

Proof. The coefficients are polynomial and hence locally Lipschitz, so a unique strong solution exists up to its explosion time. With $V = 1 + E$, discard the nonpositive term in (8) and use $M \leq 2\sqrt{nE}$ from lemma 2.1 to obtain

$$\mathcal{L}V \leq 8\gamma(T_1 + T_n)\sqrt{nE} \leq C_n V.$$

Apply Itô's formula to the process stopped on a sublevel set of V and use Gronwall's inequality. The resulting bound on the stopped expectation is uniform in the stopping level. Since V has compact sublevel sets, the usual Khasminskii stopping-time argument sends the explosion probability to zero. $\square$

3 Equilibrium consistency

Proposition 3.1 (Equal-temperature Gibbs invariance). *If $T_1 = T_n = T$, then*

$$\mu_T(dz) = Z_{n,T}^{-1} e^{-E(z)/T} dz \quad (9)$$

is an invariant probability measure for (2). This proposition does not assert uniqueness or a rate of convergence.

Proof. Normalizability follows from lemma 2.1. The Hamiltonian operator A is divergence-free and $AE = 0$, hence $A^*e^{-E/T} = 0$. Each bath operator can be written

$$B_r f = \gamma T e^{E/T} \nabla_r \cdot \left(e^{-E/T} \nabla_r f \right),$$

so it is symmetric in $L^2(\mu_T)$ and annihilates constants. Thus the weak adjoint of the full generator annihilates the density in (9). Global nonexplosion from theorem 2.3 rules out loss of mass. $\square$

4 Heat and entropy production

For bath r , heat is positive when delivered to the chain. Its Stratonovich definition and Itô form are

$$dQ_r = F_r \circ \left[-\gamma F_r dt + \sqrt{2\gamma T_r} dW_{r,t} \right], \quad (10)$$

$$= [-\gamma |F_r|^2 + \gamma T_r \Delta_r E] dt + \sqrt{2\gamma T_r} F_r \cdot dW_{r,t}. \quad (11)$$

The medium entropy transferred to the reservoirs is

$$\Sigma_t^m = - \sum_{r \in \{1,n\}} \beta_r Q_r(t), \quad \beta_r = T_r^{-1}. \quad (12)$$

Using (7), its differential is

$$d\Sigma_t^m = \sum_r [\gamma \beta_r |F_r|^2 - \gamma \Delta_r E] dt - \sum_r \sqrt{2\gamma \beta_r} F_r \cdot dW_{r,t}. \quad (13)$$

Equations (11) and (13) coincide with the continuous observable implemented in the entropy samplers.

Proposition 4.1 (Finite-time path relation under local absolute continuity). *Let the forward diffusion start from a density ρ_0 , and let the conjugate reverse experiment start from $\rho_t^R(w) = \rho_t(\Theta w)$, the time-reversed terminal density. Assume the corresponding path measures are mutually absolutely continuous on $[0, t]$. Then the path log-likelihood ratio is*

$$\Sigma_t^{\text{tot}} = \Sigma_t^{\text{m}} + \log \rho_0(Z_0) - \log \rho_t(Z_t), \quad (14)$$

because $\rho_t^R(\Theta Z_t) = \rho_t(Z_t)$. It obeys

$$\mathbb{E}_{\text{F}} e^{-\Sigma_t^{\text{tot}}} = 1 \quad (15)$$

and the forward–reverse Crooks relation.

Proof. The Hamiltonian flow is volume preserving and reversible under (4), so it contributes no path action. For each bath, the Gaussian forward/reverse increment ratio converges in the Stratonovich convention to $-\beta_r dQ_r$; equivalently this is the local-detailed-balance Radon–Nikodym calculation for an additive Langevin bath. Summing the two bath contributions and including the initial-density ratio gives (14). Equation (15) is normalization of the Radon–Nikodym derivative. Applying the path-reversal change of variables gives the forward–reverse detailed relation. $\square$

Remark 4.2 (Forward-only detailed symmetry). The exact finite-time statement is generally a forward–reverse relation. A same-ensemble formula $P_{\text{F}}(s)/P_{\text{F}}(-s) = e^s$ needs an additional self-conjugacy condition. It should not be inferred from the integral identity alone, especially when the NESS is not invariant under complex conjugation.

5 Exact tilted-generator symmetry

For a test function f , define the medium-entropy Feynman–Kac semigroup

$$P_{t,k}f(z) = \mathbb{E}_z \left[e^{-k\Sigma_t^{\text{m}}} f(Z_t) \right].$$

The standard Itô product calculation using (2) and (13) gives the tilted generator

$$\boxed{\mathcal{L}_k = A + \sum_r [\gamma T_r \Delta_r + \gamma(2k-1)F_r \cdot \nabla_r + \gamma\beta_r(k^2 - k)|F_r|^2 + \gamma k \Delta_r E]}. \quad (16)$$

Theorem 5.1 (Formal Gallavotti–Cohen identity). *On $C_c^\infty(\mathbb{R}^{2n})$, with adjoint taken relative to Lebesgue measure,*

$$\boxed{\mathcal{L}_k^* = \mathcal{T} \mathcal{L}_{1-k} \mathcal{T}}. \quad (17)$$

This identity holds for every fixed finite n , every $k \in \mathbb{R}$, and arbitrary positive endpoint temperatures.

Proof. The Hamiltonian part satisfies $A^* = -A = \mathcal{T} A \mathcal{T}$. For one bath, write $\alpha_k = \gamma(2k-1)$. Since

$$(F_r \cdot \nabla_r)^* = -F_r \cdot \nabla_r - \Delta_r E,$$

the adjoint changes the drift coefficient to $-\alpha_k = \gamma(2(1-k)-1)$ and changes the coefficient of $\Delta_r E$ from γk to

$$\gamma k - \alpha_k = \gamma(1-k).$$

Finally, $k^2 - k = (1-k)^2 - (1-k)$. The bath operator is invariant under $\mathcal{T}$ because $E \circ \Theta = E$. These identities give (17) after summing the two baths. $\square$

Assumption 5.2 (Spectral realization). *For k in a symmetric interval K with $k \in K$ iff $1-k \in K$, the closure of (16) generates a positive Feynman–Kac semigroup on a common weighted function space. The unequal-temperature process admits an invariant probability law ρ_n with the exponential moments required by that space. The long-time growth rate exists and is the isolated principal spectral value*

$$\psi_n(k) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_{\rho_n} e^{-k \Sigma_t^m}. \quad (18)$$

The domains used for k and $1-k$ are mapped into one another by $\mathcal{T}$ and adjunction.

Corollary 5.3 (Finite-chain GC symmetry). *Under assumption 5.2,*

$$\boxed{\psi_n(k) = \psi_n(1-k), \quad k \in K.} \quad (19)$$

Proof. Adjoint and $\mathcal{T}$ -conjugate closed operators have the same principal spectral value. Apply theorem 5.1 and the spectral identification in assumption 5.2. $\square$

6 The right-bath gauge used numerically

For any constant g , define

$$A_g = (-\beta_1 + g)Q_1 + (-\beta_n + g)Q_n = \Sigma^m + g(E_t - E_0).$$

The equality is exact for the continuous model. The production cloning runs use $g = \beta_1$, so

$$A_{\beta_1} = -(\beta_n - \beta_1)Q_n. \quad (20)$$

Proposition 6.1 (Coboundary similarity). *Let $G(z) = gE(z)$. On any function space for which multiplication by $e^{\pm kG}$ preserves the relevant domains, the tilted semigroups and generators for A_g and Σ^m are similar:*

$$P_{t,k}^{(g)} = e^{kG} P_{t,k} e^{-kG}, \quad \mathcal{L}_k^{(g)} = e^{kG} \mathcal{L}_k e^{-kG}. \quad (21)$$

They therefore have the same principal SCGF.

Proof. Insert $A_g = \Sigma^m + G(Z_t) - G(Z_0)$ into the definition of the tilted semigroup and move the two endpoint factors outside and inside the expectation. $\square$

Remark 6.2. Because E is unbounded, the domain condition in proposition 6.1 is a real analytic requirement, not a cosmetic technicality. The numerical decay of the endpoint difference is evidence for it but not a proof.

7 Thermodynamic-limit inheritance

Theorem 7.1 (Conditional $n \rightarrow \infty$ inheritance). *Let K be symmetric about $1/2$. Suppose (19) holds for every finite n and the pointwise limit*

$$\psi_\infty(k) = \lim_{n \rightarrow \infty} \psi_n(k) \quad (22)$$

exists and is finite for every $k \in K$. Then

$$\psi_\infty(k) = \psi_\infty(1-k), \quad k \in K. \quad (23)$$

Proof. For $k \in K$,

$$\psi_\infty(k) = \lim_n \psi_n(k) = \lim_n \psi_n(1 - k) = \psi_\infty(1 - k).$$

□

Remark 7.2 (Possible triviality). If the entropy-current scale vanishes with chain length, the unnormalized limit may be $\psi_\infty \equiv 0$. Such a limit satisfies (23) but carries no nontrivial transport information. A normalized SCGF can be studied only after its scaling and common tilt domain are fixed independently of the observed finite-size data.

8 Proof ledger and remaining work

Statement	Status	Location or missing ingredient
Energy coercivity and Gibbs normalizability	proved	lemma 2.1
Finite- n global nonexplosion	proved	theorem 2.3
Equal-temperature Gibbs invariance	proved	proposition 3.1
Heat/medium-entropy formula	proved	equations (11) and (13)
Finite-time IFT/Crooks relation	proved under path absolute continuity	proposition 4.1
Tilted-generator formal-adjoint GC identity	proved	theorem 5.1
Unequal-temperature NESS existence/uniqueness	open for this thermostat and general n	Lyapunov, irreducibility, hypoellipticity
Principal SCGF and common tilt domain	assumed	weighted Feynman–Kac spectral theory
Finite- n SCGF symmetry	conditional theorem	corollary 5.3
Right-bath/medium gauge equivalence	conditional on domains	proposition 6.1
Thermodynamic-limit symmetry	conditional theorem	theorem 7.1
Existence and nontriviality of the limit	open	uniform-in- n estimates and scaling

The next mathematical target is therefore not more algebra. It is a model-specific Lyapunov and Feynman–Kac argument for the unequal-temperature Cartesian thermostat. The present identity should be used as the symmetry step after that existence theory is supplied, following the architecture used for stochastic Markov dynamics by Lebowitz–Spohn and for boundary-driven anharmonic chains by Rey-Bellet–Thomas.

References

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